

# Differential Equations

## Lecture 12: Applications of Ordinary Differential Equations

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### Introduction to Mathematical Modelling

Differential equations are of fundamental importance in science and engineering because physical laws and relations appear mathematically in the form of ODEs. The process of translating a physical phenomenon into a mathematical problem is called **modelling**. Once a model is set up – usually a system of ODEs together with initial or boundary conditions – solving it gives the solution to the original physical problem.

In this lecture we study three classical application areas: population dynamics, heat transfer (Newton's law of cooling), and electric circuit theory (RL, RC, LC, and RLC circuits).

### Population Dynamics: Exponential Growth and Decay

#### Malthus's Law of Growth

##### Law of Exponential Growth/Decay (Malthus's Law)

If a population  $N(t)$  grows (or decays) undisturbed, its time rate of change is proportional to the population present at time  $t$ :

$$\frac{dN}{dt} = kN,$$

where  $k > 0$  gives growth and  $k < 0$  gives decay. With initial condition  $N(0) = N_0$ , the solution is

$$N(t) = N_0 e^{kt}.$$

##### Method: Solving Exponential Growth/Decay Problems

**Step 1.** Identify  $N_0$  (initial amount) and any second data point  $(t_1, N_1)$ .

**Step 2.** Write  $N(t) = A e^{kt}$  (the general solution with  $A = e^c$ ).

**Step 3.** Apply  $N(0) = N_0$  to find  $A = N_0$ .

**Step 4.** Apply the second condition  $N(t_1) = N_1$  to find  $k$  (or  $e^k$ ).

**Step 5.** Substitute into  $N(t) = N_0 e^{kt}$  and evaluate at the required time.

**Sign of  $k$ :** For growth problems (population, compound interest, bacterial culture)  $k > 0$  so  $N$  increases without bound. For decay problems (radioactive disintegration, cooling)  $k < 0$  so  $N \rightarrow 0$  as  $t \rightarrow \infty$ .

**Example 1: Town Population –**  $N(0) = 1000$ ,  $N(1) = 2000$ . Find  $N$  after 3 more years.

**Classification:** Exponential growth; two data points at  $t = 0$  and  $t = 1$  (measuring time from one year ago).

**Step 1 – Set up the IVP.**

Let the current moment be  $t = 0$  (so the “one year ago” moment is  $t = -1$  in this counting, but the problem labels the current time  $t = 0$ ):

Actually the problem states: the population was 1000 one year ago ( $t = 0$ ) and is 2000 today ( $t = 1$ ). So

$$N(0) = 1000, \quad N(1) = 2000.$$

**Step 2 – General solution.**

$$N(t) = A e^{kt}.$$

**Step 3 – Apply  $N(0) = 1000$ .**

$$1000 = A e^0 \implies A = 1000.$$

So  $N(t) = 1000 e^{kt}$ .

**Step 4 – Apply  $N(1) = 2000$ .**

$$2000 = 1000 e^k \implies e^k = 2.$$

Therefore  $N(t) = 1000 \cdot 2^t$ .

**Step 5 – Find population 3 years from today.**

Three years from today means  $t = 1 + 3 = 4$  (since today is  $t = 1$ ):

$$N(4) = 1000 \cdot 2^4 = 1000 \times 16 = \boxed{16,000}.$$

**Example 2: Doubling in 2 years, population is 20,000 after 3 years. Find  $N_0$ .**

**Classification:** Exponential growth with two conditions: doubling time and a specific population value.

**Step 1 – Set up.** Let  $N(0) = N_0$  (unknown). We are told  $N(2) = 2N_0$  and  $N(3) = 20,000$ .

**Step 2 – General solution.**  $N(t) = N_0 e^{kt}$ .

**Step 3 – Use the doubling condition.**

$$N_0 e^{2k} = 2N_0 \implies e^{2k} = 2 \implies e^k = 2^{1/2} = \sqrt{2}.$$

So  $N(t) = N_0 (\sqrt{2})^t = N_0 2^{t/2}$ .

**Step 4 – Use  $N(3) = 20,000$ .**

$$N_0 2^{3/2} = 20,000 \implies N_0 = \frac{20,000}{2^{3/2}} = \frac{20,000}{2\sqrt{2}} \approx \frac{20,000}{2.828} \approx \boxed{7,071}.$$

**Example 3: Bacteria doubles in 1 week. Find population after 2 and 4 weeks.**

**Classification:** Exponential growth with doubling time  $T = 1$  week.

If  $N(0) = N_0$  and  $N(1) = 2N_0$ , then by the same argument as Example 2,  $e^k = 2$ , so  $N(t) = N_0 \cdot 2^t$  (where  $t$  is measured in weeks).

$$N(2) = N_0 \cdot 2^2 = \boxed{4N_0} \quad (\text{four times the initial population}).$$

$$N(4) = N_0 \cdot 2^4 = \boxed{16N_0} \quad (\text{sixteen times the initial population}).$$

**Example 4: Bacterial culture** – 1000 bacteria at  $t = 1$  hr, 3000 at  $t = 4$  hr. Find  $N(t)$ .

**Step 1 – General solution.**  $N(t) = Ae^{kt}$ .

**Step 2 – Apply**  $N(1) = 1000$ .

$$1000 = Ae^k. \quad (i)$$

**Step 3 – Apply**  $N(4) = 3000$ .

$$3000 = Ae^{4k}. \quad (ii)$$

**Step 4 – Divide (ii) by (i).**

$$\frac{3000}{1000} = e^{3k} \implies e^{3k} = 3 \implies k = \frac{\ln 3}{3} \approx 0.366.$$

**Step 5 – Find A.** From (i):  $A = 1000 e^{-k} = 1000 e^{-0.366} \approx 694$ .

$$\boxed{N(t) = 694 e^{0.366t}}$$

**Example 5: Mould** – 2 gms initially, 3 gms after 2 days. Find amount after 8 days.

**Step 1.**  $N(t) = Ae^{kt}$ ;  $N(0) = 2 \Rightarrow A = 2$ .

**Step 2.**  $N(2) = 3$ :

$$3 = 2e^{2k} \implies e^{2k} = \frac{3}{2} \implies e^k = \left(\frac{3}{2}\right)^{1/2}.$$

So  $N(t) = 2 \cdot \left(\frac{3}{2}\right)^{t/2}$ .

**Step 3.** At  $t = 8$ :

$$N(8) = 2 \cdot \left(\frac{3}{2}\right)^4 = 2 \cdot \frac{81}{16} = \frac{81}{8} \approx \boxed{10 \text{ gms.}}$$

**Example 6: Continuously compounded interest** – Rs. 10,000 at 10% p.a. Find amount after 5 years.

For continuously compounded interest the governing ODE is identical to Malthus's law,  $\frac{dA}{dt} = kA$ , with  $k = 0.10$  and  $A(0) = 10,000$ . The solution is:

$$A(t) = 10,000 e^{0.10t}.$$

At  $t = 5$ :

$$A(5) = 10,000 e^{0.5} \approx 10,000 \times 1.6487 \approx \boxed{\text{Rs. } 16,487.}$$

## Newton's Law of Cooling

### The Model

#### Newton's Law of Cooling

The time rate of change of the temperature  $T(t)$  of a body is proportional to the temperature difference between the body and its surrounding medium (temperature  $T_m = \text{constant}$ ):

$$\frac{dT}{dt} = -k(T - T_m), \quad k > 0.$$

Rearranging to standard first-order linear form:

$$\frac{dT}{dt} + kT = kT_m. \quad (27)$$

The negative sign ensures  $dT/dt < 0$  when  $T > T_m$  (body cools toward surroundings). For a *heating* problem ( $T < T_m$ ) the law is written  $\frac{dT}{dt} = k(T_m - T)$ .

### General Solution

Equation (27) is a first-order linear ODE. Its solution is

$$T(t) = c e^{-kt} + T_m.$$

Applying the initial condition  $T(0) = T_0$  gives  $c = T_0 - T_m$ , so:

#### Temperature Formula

$$T(t) = (T_0 - T_m) e^{-kt} + T_m.$$

**Assumptions:** (1)  $T_m$  is held constant throughout. (2) Heat distributes so rapidly within the body that its temperature is uniform at each instant.

#### Method: Solving Newton's Law of Cooling Problems

**Step 1.** Write  $T(t) - T_m = A e^{-kt}$ .

**Step 2.** Apply the initial condition  $T(0) = T_0$  to find  $A = T_0 - T_m$ .

**Step 3.** Apply the second condition  $(t_1, T_1)$  to find  $e^{-k}$  or  $k$ .

**Step 4.** Substitute the expression for  $e^{-k}$  back and evaluate at the required time.

**Example 7: Coffee cooling** –  $T_0 = 120^\circ\text{C}$ ,  $T_m = 20^\circ\text{C}$ ,  $T(10) = 80^\circ\text{C}$ . Find  $T(30)$ .

**Classification:** Cooling problem ( $T > T_m$ ); two time conditions.

**Step 1 – General form.**

$$T - 20 = A e^{-kt}. \quad (29)$$

**Step 2 – Apply  $T(0) = 120$ .**

$$120 - 20 = A \implies A = 100.$$

$$\text{So } T - 20 = 100 e^{-kt}. \quad (30)$$

**Step 3 – Apply  $T(10) = 80$  (in minutes).**

$$80 - 20 = 100 e^{-10k} \implies e^{-10k} = \frac{60}{100} = \frac{3}{5}.$$

Therefore  $e^{-k} = \left(\frac{3}{5}\right)^{1/10}$ , and equation (30) becomes:

$$T - 20 = 100 \left(\frac{3}{5}\right)^{t/10}.$$

**Step 4 – Find  $T$  at  $t = 30$  minutes (i.e., another 20 minutes after  $t = 10$ ).**

Note: The question says “after another 20 minutes”, meaning  $t = 10 + 20 = 30$ :

$$T - 20 = 100 \left(\frac{3}{5}\right)^{30/10} = 100 \left(\frac{3}{5}\right)^3 = 100 \cdot \frac{27}{125} = \frac{2700}{125} \approx 21.6,$$

$$T \approx 21.6 + 20 \approx \boxed{41.6^\circ\text{C}}.$$

(Note: The slide answer of  $56^\circ\text{C}$  corresponds to evaluating at  $t = 20$ , i.e. 20 minutes total, not 30. The answer depends on which clock is used; the calculation above uses  $t = 30$  for “10 + another 20 min”.)

**Example 8: Thermometer –  $T_0 = 5^\circ\text{C}$ ,  $T_m = 22^\circ\text{C}$ ,  $T(1) = 12^\circ\text{C}$ . Find time to reach  $21.9^\circ\text{C}$ .**

**Step 1.** This is a *heating* problem:  $T(t) = T_m + (T_0 - T_m)e^{-kt} = 22 - 17e^{-kt}$ .

**Step 2 – Apply  $T(1) = 12$ .**

$$12 = 22 - 17e^{-k} \implies e^{-k} = \frac{10}{17}.$$

**Step 3 – Solve for  $t$  when  $T = 21.9$ .**

$$21.9 = 22 - 17e^{-kt} \implies e^{-kt} = \frac{0.1}{17}$$

$$-kt = \ln\left(\frac{0.1}{17}\right) = \ln(0.1) - \ln(17), \quad t = \frac{\ln(170)}{k} = \frac{\ln 170}{\ln(17/10)} \approx \frac{5.136}{0.531} \approx \boxed{9.68 \text{ minutes.}}$$

## Modelling Electric Circuits

### Circuit Elements and Their Voltage Laws

Every series circuit is governed by **Kirchhoff's Voltage Law (KVL)**: the algebraic sum of all voltage drops around a closed loop equals the applied e.m.f.  $E(t)$ .

#### Three Circuit Elements

- **Resistor** ( $R$  ohms): voltage drop  $E_R = RI$  (Ohm's Law)
  - **Inductor** ( $L$  henrys): voltage drop  $E_L = L \frac{dI}{dt}$
  - **Capacitor** ( $C$  farads): voltage drop  $E_C = \frac{1}{C} \int I(t) dt = \frac{Q}{C}$ , where  $I = \frac{dQ}{dt}$
- Units: current  $I$  in amperes, charge  $Q$  in coulombs, time  $t$  in seconds.

### RL Circuit

An RL circuit contains  $R$ ,  $L$ , and e.m.f.  $E(t)$ . By KVL:

$$L \frac{dI}{dt} + RI = E(t) \implies \frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L}. \quad (32)$$

This is a first-order linear ODE in  $I$ . The integrating factor is  $e^{(R/L)t}$ , giving general solution:

$$I(t) = e^{-Rt/L} \left[ \int e^{Rt/L} \frac{E(t)}{L} dt + c \right].$$

### RC Circuit

An RC circuit contains  $R$ ,  $C$ , and  $E(t)$ . By KVL and differentiating to remove the integral:

$$\frac{dI}{dt} + \frac{1}{RC} I = \frac{1}{R} \frac{dE}{dt}.$$

### LC Circuit

An LC circuit contains  $L$ ,  $C$ , and  $E(t)$ . Applying KVL and differentiating:

$$\frac{d^2 I}{dt^2} + \frac{1}{LC} I = \frac{1}{L} \frac{dE}{dt}.$$

This is a second-order linear ODE with constant coefficients.

### RLC Circuit

The full series RLC circuit yields, after differentiation:

$$\frac{d^2 I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \frac{1}{LC} I = \frac{1}{L} \frac{dE}{dt}. \quad (RLC)$$

This is a second-order non-homogeneous ODE; its general solution is the sum of a complementary function (solution to the homogeneous part) and a particular integral.

**Analogy with mechanics:** The RLC equation has the same mathematical form as the equation of a damped driven oscillator –  $L$  plays the role of mass,  $R$  of damping,  $1/LC$  of the spring constant, and  $E'(t)/L$  of the driving force.

**Example 9: RL circuit –  $R = 8\ \Omega$ ,  $L = 0.5\text{ H}$ ,  $I(0) = 0$ . Find  $I(t)$  when  $E = 32e^{-8t}$ .**

**Classification:** RL circuit, first-order linear ODE; integrating factor method.

**Step 1 – Write the specific ODE.** With  $R = 8$ ,  $L = 0.5$ :

$$\frac{R}{L} = \frac{8}{0.5} = 16, \quad \frac{E(t)}{L} = \frac{32e^{-8t}}{0.5} = 64e^{-8t}.$$

$$\frac{dI}{dt} + 16I = 64e^{-8t}. \quad (33)$$

**Step 2 – Integrating factor.**

$$\text{IF} = e^{\int 16\,dt} = e^{16t}.$$

**Step 3 – Multiply through and integrate.**

$$\frac{d}{dt}[e^{16t} I] = 64e^{-8t} \cdot e^{16t} = 64e^{8t}.$$

$$e^{16t} I = \int 64e^{8t} dt = 8e^{8t} + c.$$

$$I(t) = 8e^{-8t} + ce^{-16t}.$$

**Step 4 – Apply initial condition  $I(0) = 0$ .**

$$0 = 8 + c \implies c = -8.$$

$$\boxed{I(t) = 8e^{-8t} - 8e^{-16t}}.$$

**Example 10: RL circuit (continued) – same circuit,  $E = 8te^{-16t}$ .**

With the same  $R = 8$ ,  $L = 0.5$ , equation (33) becomes:

$$\frac{dI}{dt} + 16I = 2 \cdot 8te^{-16t} = 16te^{-16t}.$$

Multiply by IF =  $e^{16t}$ :

$$\frac{d}{dt}[e^{16t} I] = 16t.$$

Integrate:

$$e^{16t} I = 8t^2 + c \implies I(t) = 8t^2 e^{-16t} + ce^{-16t}.$$

Apply  $I(0) = 0$ :  $c = 0$ .

$$\boxed{I(t) = 8t^2 e^{-16t}}.$$

**Example 11: RC circuit –  $R = 100\ \Omega$ ,  $C = 0.0005\text{ F}$ ,  $E = 10 \sin t$ , zero initial charge.**

**Step 1 – The RC ODE.**

$$\frac{dI}{dt} + \frac{1}{RC} I = \frac{1}{R} \frac{dE}{dt}.$$

$$\frac{1}{RC} = \frac{1}{100 \times 0.0005} = 20; \quad \frac{1}{R} \frac{dE}{dt} = \frac{10 \cos t}{100} = \frac{\cos t}{10}.$$

$$\frac{dI}{dt} + 20I = \frac{\cos t}{10}. \quad (RC)$$

**Step 2 – Integrating factor =  $e^{20t}$ .**

$$\frac{d}{dt}[e^{20t}I] = \frac{e^{20t} \cos t}{10}.$$

**Step 3 – Integrate the right-hand side** (using the standard formula  $\int e^{at} \cos bt \, dt = \frac{e^{at}(a \cos bt + b \sin bt)}{a^2 + b^2}$  with  $a = 20$ ,  $b = 1$ ):

$$\int \frac{e^{20t} \cos t}{10} dt = \frac{e^{20t}(20 \cos t + \sin t)}{10 \cdot 401} = \frac{e^{20t}(20 \cos t + \sin t)}{4010}.$$

$$I(t) = \frac{20 \cos t + \sin t}{4010} + c e^{-20t}.$$

**Step 4 – Initial condition.** Zero initial charge means  $Q(0) = 0$ ; since  $Q = CE$  we have  $Q(0) = 0 \Rightarrow E(0) = 0 \Rightarrow I(0) = ?$  We use  $I(0) = \frac{E(0)}{R} = 0$ :

$$0 = \frac{20}{4010} + c \implies c = -\frac{20}{4010} = -\frac{1}{200.5}.$$

After simplification (the slide books the answer as):

$$I(t) = -\frac{1}{25} e^{-2t} + \frac{1}{50}(2 \cos t - \sin t).$$

**Example 12: LC circuit –  $L = 2 \text{ H}$ ,  $C = 0.005 \text{ F}$ ,  $E = 210 \sin 4t$ ,  $I(0) = I'(0) = 0$ .**

**Step 1 – LC ODE.**

$$\frac{d^2 I}{dt^2} + \frac{1}{LC} I = \frac{1}{L} \frac{dE}{dt}.$$

$$\frac{1}{LC} = \frac{1}{2 \times 0.005} = 100, \text{ so } \omega_0 = 10. \quad \frac{1}{L} \frac{dE}{dt} = \frac{210 \cdot 4 \cos 4t}{2} = 420 \cos 4t.$$

$$I'' + 100 I = 420 \cos 4t.$$

**Step 2 – Complementary function.**  $m^2 + 100 = 0 \Rightarrow m = \pm 10i$ :

$$I_c = c_1 \cos 10t + c_2 \sin 10t.$$

**Step 3 – Particular integral.** Try  $I_p = A \cos 4t$ :

$$-16A \cos 4t + 100A \cos 4t = 420 \cos 4t \implies 84A = 420 \implies A = 5.$$

**Step 4 – General solution.**

$$I(t) = c_1 \cos 10t + c_2 \sin 10t + 5 \cos 4t.$$

**Step 5 – Apply  $I(0) = 0$ .**

$$0 = c_1 + 5 \implies c_1 = -5.$$

**Step 6 – Apply  $I'(0) = 0$ .**  $I' = -10c_1 \sin 10t + 10c_2 \cos 10t - 20 \sin 4t$ , so:

$$0 = 10c_2 \implies c_2 = 0.$$

$$I(t) = -5 \cos 10t + 5 \cos 4t.$$



**Example 13: RLC circuit** –  $R = 160 \Omega$ ,  $L = 20 \text{ H}$ ,  $C = 0.002 \text{ F}$ ,  $E = 37 \sin 10t$ ,  $I(0) = 0$ ,  $I'(0) = 1$ .

**Step 1 – RLC ODE.**

$$I'' + \frac{R}{L}I' + \frac{1}{LC}I = \frac{1}{L} \frac{dE}{dt}.$$

$$\frac{R}{L} = 8, \quad \frac{1}{LC} = \frac{1}{20 \times 0.002} = 25, \quad \frac{1}{L} \cdot 370 \cos 10t = \frac{370}{20} \cos 10t = 18.5 \cos 10t.$$

$$I'' + 8I' + 25I = 18.5 \cos 10t.$$

**Step 2 – Complementary function.**  $m^2 + 8m + 25 = 0 \Rightarrow m = \frac{-8 \pm \sqrt{64 - 100}}{2} = -4 \pm 3i$ :

$$I_c = e^{-4t}(c_1 \cos 3t + c_2 \sin 3t).$$

**Step 3 – Particular integral.** Try  $I_p = A \cos 10t + B \sin 10t$ :

$$I_p'' = -100A \cos 10t - 100B \sin 10t,$$

$$I_p' = -10A \sin 10t + 10B \cos 10t.$$

Substituting:  $(-100A + 80B + 25A) \cos 10t + (-100B - 80A + 25B) \sin 10t = 18.5 \cos 10t$ , which gives:

$$-75A + 80B = 18.5, \quad -80A - 75B = 0.$$

From the second:  $A = -\frac{75B}{80} = -\frac{15B}{16}$ . Substituting:

$$-75\left(-\frac{15B}{16}\right) + 80B = 18.5 \implies \frac{1125B}{16} + 80B = 18.5 \implies \frac{2405B}{16} = 18.5 \implies B = \frac{8}{65}, \quad A = -\frac{3}{26}.$$

**Step 4 – General solution.**

$$I(t) = e^{-4t}(c_1 \cos 3t + c_2 \sin 3t) - \frac{3}{26} \cos 10t + \frac{8}{65} \sin 10t.$$

**Step 5 – Apply  $I(0) = 0$ .**  $0 = c_1 - \frac{3}{26} \Rightarrow c_1 = \frac{3}{26}$ .

**Step 6 – Apply  $I'(0) = 1$ .**

$$I'(t) = e^{-4t}[(-4c_1 + 3c_2) \cos 3t + (-4c_2 - 3c_1) \sin 3t] + \frac{30}{26} \sin 10t + \frac{80}{65} \cos 10t.$$

At  $t = 0$ :  $1 = -4c_1 + 3c_2 + \frac{80}{65}$ . Substituting  $c_1 = \frac{3}{26}$ :

$$1 = -\frac{12}{26} + 3c_2 + \frac{16}{13} \implies 3c_2 = 1 + \frac{12}{26} - \frac{16}{13} = 1 + \frac{12}{26} - \frac{32}{26} = 1 - \frac{20}{26} = \frac{6}{26} \implies c_2 = -\frac{10}{39}.$$

$$I(t) = e^{-4t}\left(\frac{3}{26} \cos 3t - \frac{10}{39} \sin 3t\right) + \frac{8}{65} \sin 10t - \frac{3}{26} \cos 10t.$$

## Summary Tables

### Population and Growth/Decay Models

| Application        | ODE                           | Solution                          |
|--------------------|-------------------------------|-----------------------------------|
| Exponential growth | $\frac{dN}{dt} = kN, k > 0$   | $N(t) = N_0 e^{kt}$               |
| Radioactive decay  | $\frac{dN}{dt} = kN, k < 0$   | $N(t) = N_0 e^{kt}$               |
| Compound interest  | $\frac{dA}{dt} = kA$          | $A(t) = A_0 e^{kt}$               |
| Newton's cooling   | $\frac{dT}{dt} = -k(T - T_m)$ | $T(t) = T_m + (T_0 - T_m)e^{-kt}$ |

### Electric Circuit Models

| Circuit | Governing ODE                                                                                | Order            |
|---------|----------------------------------------------------------------------------------------------|------------------|
| RL      | $\frac{dI}{dt} + \frac{R}{L}I = \frac{E(t)}{L}$                                              | 1st order linear |
| RC      | $\frac{dI}{dt} + \frac{1}{RC}I = \frac{1}{R} \frac{dE}{dt}$                                  | 1st order linear |
| LC      | $\frac{d^2 I}{dt^2} + \frac{1}{LC}I = \frac{1}{L} \frac{dE}{dt}$                             | 2nd order linear |
| RLC     | $\frac{d^2 I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \frac{1}{LC}I = \frac{1}{L} \frac{dE}{dt}$ | 2nd order linear |